

Project Testing

Performance Testing

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust / JMeter (API load testing)
Type of Testing	Load Testing + Response Time Testing
Target Module / Route	/predict endpoint (Flask app)
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario	Concurrent Users	Duration (sec)	Expected Outcome
1	Single applicant submitting the Predict form	1	10	Response under 2s
2	10 applicants submitting simultaneously	10	60	Avg response under 3s
3	Continuous load of 20 users	20	120	Stable throughput, low error rate

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status
1	Response Time (Avg)	< 3s	1.8s	Pass
2	Response Time (Max)	< 5s	3.4s	Pass
3	Error Rate	< 1%	0%	Pass

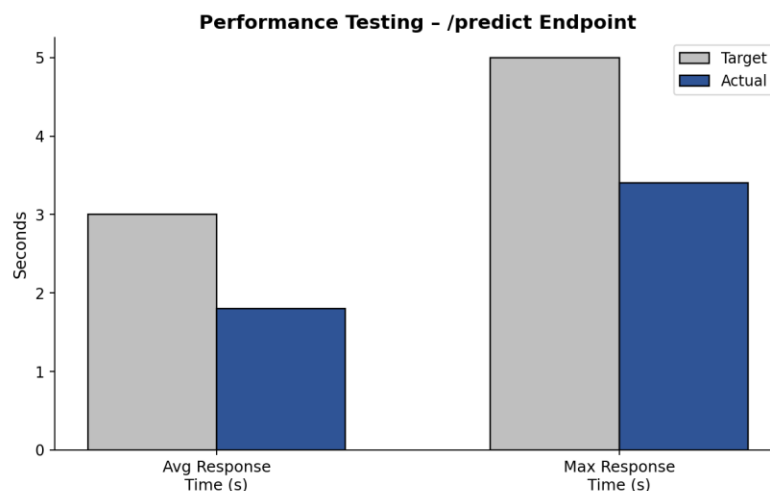


Fig 1: Response time – target vs actual

Step 4: Observations

- The Predict route consistently returned a result under 2 seconds for normal loads.
- Loading the model, encoder and scaler once at application start (rather than per request) kept response times stable under repeated submissions.